



Sensors for biomanufacturing process development: facilitating the shift from batch to continuous manufacturing

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The biopharmaceutical industry has traditionally utilized batch manufacturing strategies for production, but continuous manufacturing has been explored as an alternative production method in recent years because of the potential financial and quality benefits. The complex nature of this manufacturing method requires robust online process analytical technology (PAT) that can effectively characterize continuous processes at the development scale (0.1–100 mL for process scouting devices (PSDs) and 100 mL–10 L for bench-scale development) as well as sensors that can monitor and control them at the commercial scale (hundreds to thousands of liters). Both established and new bioprocessing sensor technologies for characterizing and monitoring batch and continuous bioprocesses are explored in this review and are evaluated for their potential use in continuous bioprocess development and in commercial-scale production. Not all of these sensors can be integrated with control systems for end-to-end continuous bioprocesses at the commercial scale, but they can currently be used to gain a better understanding of new continuous processes at the development scale in order to make the scale-up process more data-driven, improving process understanding.

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strategy is used in many industries, such as steel casting and chemical manufacturing [1], but it is still uncommon in the biopharmaceutical industry, which generally employs batch manufacturing [2]. There are many issues with commercial-scale continuous biomanufacturing that are not observed in or are not relevant to other industries. Maintaining sterility is difficult when clean-in-place and steam-in-place procedures can't be conducted as frequently as they are in batch manufacturing. Identifying the cause of specific deviations during production is more difficult when the product flows continuously through the different unit operations [2]. Process development can be expensive and time-consuming [3]. There are some generic problems that have been observed in continuous manufacturing of biopharmaceuticals, and there are likely more product-specific issues that companies would need to address if they want to utilize this production strategy [2]. Despite these concerns, widespread use of this type of manufacturing in the biopharmaceutical industry could have significant product quality and financial benefits. Integrating the unit operations for continuous flow has been shown to decrease processing times, improve production efficiency, and reduce the size of equipment and operational costs [4].

However, online process analytical technology (PAT) that is more robust than what is currently used in batch manufacturing is needed to effectively monitor the continuous manufacturing process and communicate performance changes to process control equipment. Local control of individual unit operations is not sufficient for this production strategy because all parts of the process must be closely integrated. Some critical quality attributes (CQAs) that must be monitored include levels of bioburden, host cell protein (HCP), protein aggregates, and overall product concentration [1,4]. Global process control includes coordinating flow rates between unit operations through feedback loops, managing interruptions in the process due to deviations or other unexpected events, and automatically adjusting operating parameters in response to changes in process performance [3]. This more extensive level of control is generally not implemented in batch manufacturing.

Introduction

Continuous manufacturing is a process defined by a constant flow of material production. This production

PAT is also needed to monitor bioprocesses at the development scale. Online monitoring of cell culture

parameters in process scouting devices (PSDs) in the range of 0.1–100 mL is especially crucial for effective process characterization and scale-up, but very few robust real-time sensors exist at this scale compared to other levels of process development [5]. Many companies have already implemented continuous manufacturing upstream processes and equipment, but PSDs are useful for scale-down studies for process characterization [5]. PSDs and bench-scale vessels (100 mL–10 L) will be especially important as upstream and downstream processes become more integrated in continuous manufacturing. Robust sensors for this scale of development support the implementation of Quality by Design (QbD) principles in the biopharmaceutical industry by improving process understanding for earlier characterization of the CQAs of the product being manufactured [6,7].

Here, we explore some established sensors and emerging ones for monitoring process parameters at the development scale (and potentially larger scales) that are important in both batch and continuous bioprocessing, such as pH, ammonia, and glucose. Although these sensors are not fully equipped to monitor process variables specific to continuous manufacturing, the technology discussed here is useful for process characterization in any type of biomanufacturing and can be adapted for monitoring of continuous manufacturing parameters at larger scales in future work.

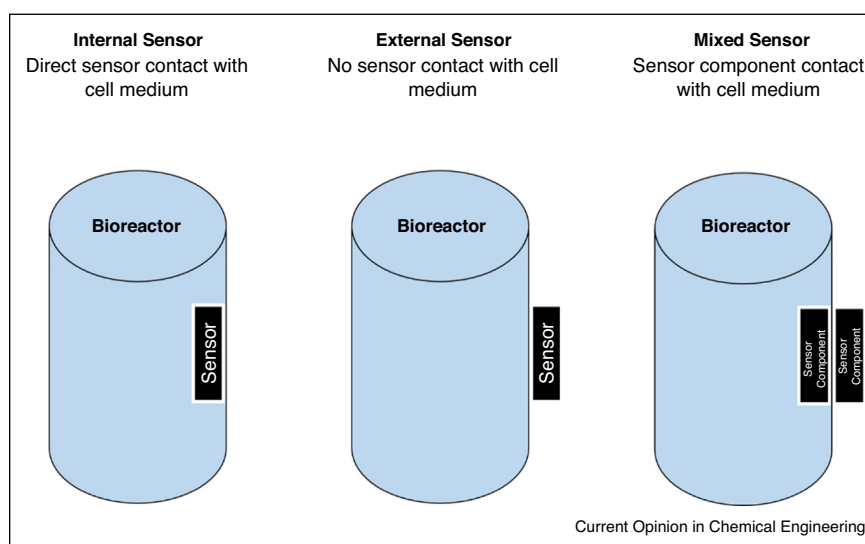
Types of bioprocessing sensors

Bioprocessing sensors can be categorized based on their level of contact with the cells and culture medium in the cell culture vessel. Internal sensors are located inside of

the vessel and directly contact the cell culture. The sensor is generally coated or protected by a biocompatible material that prevents degradation of sensor electronics and biofouling. External sensors are located outside of the vessel and have no contact with the cell culture. A sample may need to be removed from the main vessel through a closed circulation loop, but the sensor never makes direct contact with the cells. Mixed sensors have components inside and outside of the cell culture vessel. The components that contact the cells may or may not be part of the sensor itself (e.g. a sampling probe for extracting carbon dioxide from the cell culture and transporting it to the sensor sitting outside of the vessel). These designs are shown in Figure 1.

In general, the operating mechanism of most bioprocessing sensors can be defined as electrochemical or optical [8^{••}]. Electrochemical sensors measure current induced by oxidation or reduction of the target agent (amperometric) or measure the agent's induced equilibrium potential compared to a reference electrode that correlates with the concentration of the agent (potentiometric) [8^{••}]. These types of sensors usually require some level of contact with the sample. Optical sensors measure shift in absorbance, fluorescence signal emission, phase angle, and other properties related to light using indicator dyes that can be correlated with the relative concentration of the analyte. Optical sensors are becoming popular for bioprocess monitoring because many of them are capable of conducting measurements with little to no direct contact with the cell culture. Their limited long-term stability has restricted widespread use in the industry, but sensors with disposable components have been developed to circumvent this issue [8^{••}].

Figure 1



Categories of bioprocessing sensors based on level of sensor contact with cell medium.

Novel sensors for detection of environmental parameters in cell culture, including pH, dissolved oxygen (DO), and dissolved carbon dioxide (DCO₂) are outlined in Table 1. Sensors for monitoring several organic compounds, ammonia, protein product, and biomass are outlined in Table 2. All of these sensors are also evaluated for

their possible use in continuous bioprocess development at the PSD scale and bench scale, and the potential for them to eventually be used in commercial-scale production is also considered. Each sensor is assessed a score based on these evaluations, and all of the assessments are presented in the two tables.

Table 1

Sensors for cell culture environmental parameters (pH, DO and DCO₂). Traditional sensors for each analyte are italicized. Each sensor is ranked for its potential use in continuous manufacturing development based on size, robustness, and stability.

Analyte	Position	Type	Sensor Name/Description	Potential Use for Cont. Manufacturing	Rank (1–5)	References
pH	Internal	Electrochemical	<i>Glass pH sensors</i> – equilibrium potential between glass membrane and reference electrode is measured	<i>High accuracy and good stability. Difficult to miniaturize for PSD scale development.</i>	4	Demuth et al. [8**]
	Internal	Electrochemical	ISFETS^a and Capacitive EIS^b sensors – semiconductor-based sensors with a pH-sensitive transduction material	Can be miniaturized for small-scale development. Long-term stability and potential for drift need to be evaluated.	3	Bäcker et al. [10], Demuth et al. [8**], Krommenhoek et al. [12]
	Internal	Electrochemical	Voltammetric sensors – measures change in peak potential of pH-sensitive, redox active compound	Better long-term stability compared with potentiometric sensors. Not yet commercialized.	4	Demuth et al. [8**], Lafitte et al. [40]
	Internal	Electrochemical	RFID^c sensors – coiled iridium/iridium oxide pH sensor with reference electrode. Shift in pH causes shift in circuit's resonance frequency	Wireless design is simple and allows for use in PSDs and other vessel scales. Minimal long-term drift. Needs to be re-calibrated if temperature changes.	5	Bhadra et al. [11], Demuth et al. [8**]
	External	Optical	Optical sensor measurement using change in color of phenol-red cell culture medium with sterilizable, modular parts	Sensor itself can only be used with phenol-red medium, but modular sampling system could be used with other optical pH sensors for external measurements.	3	Kattipparambil Rajan et al. [15]
	Mixed	Optical	Ratiometric pH patches – ratio of emission wavelength from two colors of LED lights through fluorescent dye in polymer matrix correlated with pH	Useful for PSD-scale development, but application for larger scales is limited due to small size and fixed position. Durability for long-term continuous bioprocesses may be an issue.	4	Ge et al. [14], Sardesai et al. [41]
pH, DO ^d	Internal	Electrochemical	PATsule – pill-shaped sensor consisting of semiconductor chip and coated with polyether ether ketone (PEEK). Floats in culture medium and wirelessly transmits process data to computer	Highly suitable for development and production at different scales because of long-term stability from PEEK coating and non-static position in the reactor that provides robust dataset.	5	O'Mara et al. [13]
DCO ₂ ^e	Internal	Electrochemical	Severinghaus principle – change in pH of solution in electrode as result of CO ₂ diffusion into solution is measured as potential difference	<i>Good accuracy, but long-term stability is questionable. Difficult to miniaturize. Relatively slow response time.</i>	3	Dixon and Kell [42], Severinghaus & Bradley [22]

Table 1 (Continued)

Analyte	Position	Type	Sensor Name/Description	Potential Use for Cont. Manufacturing	Rank (1–5)	References
DO	Internal	Electrochemical	Clark sensors – amperometric measurement system based on oxygen reduction on cathode surface	Good long-term stability, but difficult to miniaturize. Frequent maintenance required. DO consumption may affect small-scale culture.	3	Demuth <i>et al.</i> [8**]
	External	Optical	Optical patch sensors placed on outside of vessel wall on gas-permeable membrane – Oxygen diffuses across the membrane to the patch	Limited for use with slow growing cell cultures due to long response time. Could be used in disposable bag reactors in the future.	2	Gupta <i>et al.</i> [20]
	Mixed	Optical	DO patch sensors – change in phase angle between excitation light and fluorescence emission through fluorescent dye in polymer matrix correlates with DO concentration	Useful for PSD-scale development, but application for larger scales is limited due to small size and fixed position. Durability for long-term continuous bioprocesses may be an issue.	4	Sardesai <i>et al.</i> [41], Tolosa <i>et al.</i> [19]
	Mixed	Optical	Nanoparticles for oxygen tension reading – optical measurements of oxygen-sensitive nanoparticles with fluorescent dye in shaker flask cell culture medium	May be more accurate than patch sensors for shake flask culture. Potential use in early-stage cell line development and media optimization.	3	Flitsch <i>et al.</i> [21*]
DO, DCO ₂	Mixed	Electrochemical	Circulation direct monitoring and sampling system (CDMSS) – system that continuously extracts and measures liquid and gas-phase samples	Potential use in early-stage cell line development and media optimization in shake flasks.	3	Takahashi <i>et al.</i> [16*]
	Mixed	Optical	Rate-based measurements – initial diffusion rate of O ₂ /CO ₂ across polymer sampling loop into closed system connected to external sensor is measured and correlated with liquid-phase O ₂ /CO ₂ concentration	Useful for PSD-scale development and potentially other scales of cont. bioprocess development. Further assessment is needed for long-term stability.	4	Chatterjee <i>et al.</i> [17]

^a Ion-Sensitive Field-Effect Transistors.
^b Electrolyte-Insulator-Semiconductors.
^c Radio-Frequency Identification.
^d Dissolved oxygen.
^e Dissolved carbon dioxide.

pH sensors

Solid-state pH sensor technology, including ion-sensitive field-effect transistors (ISFETs), capacitive electrolyte-insulator-semiconductors (EIS), voltammetric cells, and radio-frequency identification (RFID), are some of the newest types of electrochemical sensors that have emerged in recent years. Various advantages that they have over traditional glass pH sensors for process development are long-term stability, dry storage, and small size that allows them to be used in minibioreactors and other

PSDs [8**,9]. Different types of these sensors have been successfully used to monitor cell culture processes [10–12].

Another unique electrochemical pH sensor that has been recently developed is the Process Analytical Technology Sensor Capsule (PATsule), a pill-shaped sensor slightly bigger than a quarter that can float in cell culture and wirelessly transmit process data, including pH and dissolved oxygen. Unlike most sensors, the position of this

Table 2

Sensors for organic compounds (glucose, glutamine, and lactate) as well as cell biomass and by-products (ammonia and protein product). Traditional sensors for each analyte are italicized. Each sensor is ranked for its potential use in continuous manufacturing development based on size, robustness, and stability.

Analyte	Position	Type	Sensor Name/Description	Potential Use for Cont. Manufacturing	Rank (1–5)	References
Ammonia, Glucose, Glutamate, Glutamine, Lactate, Biomass	External	Electrochemical	<i>YSI 2900 Analyzer, BioProfile FLEX – amperometric biosensor</i>	<i>Gold standard, off-line metabolite analyzer. Provides reliable measurements, but less useful for continuous manufacturing because it is not capable of providing continuous online measurements as a standalone device.</i>	3	Derfus <i>et al.</i> [43], Schlueter <i>et al.</i> [44]
Ammonia, Glucose, Glutamate, Glutamine, Lactate, Protein Product, Biomass	External	Optical	Raman spectroscopy – detects inelastic scattering of light from a laser illumination source, providing molecular data of the excited molecules in the sample	Non-invasive and nondestructive analysis (sensor does not consume analyte) of analytes with low concentration detection limits. However, the detection needs very sensitive and highly optimized instrumentation. For larger scale bioprocesses, the weak signal may require very expensive machinery to detect variations in analyte concentrations. The weak signal has a high susceptibility to interference.	3	Abu-Absi <i>et al.</i> [31], Esmonde-White <i>et al.</i> [34*], Rowland-Jones <i>et al.</i> [45**]
Ammonia, Glucose, Protein Product, Biomass	Mixed	Optical	Fluorescence spectroscopy soft sensor – can predict key cell culture attributes in-between at situ samples	Soft sensors are useful for predictive modeling of complex biological systems. Predictive modeling is an advantageous method for complex systems but is not truly a reflection of real-time analyte concentrations. Fluorescence spectroscopy-based measurements can be at risk for interference depending on the sensor used.	4	Esmonde-White <i>et al.</i> [34*], Luttmann <i>et al.</i> [46], Ohadi <i>et al.</i> [33]
Glucose	External	Electrochemical	In-situ glucose electrochemical sensor and Complimentary Metal-Oxide-Semiconductor (CMOS) potentiostat with a two-layer membrane for interference elimination and wide detection range	Membranes help eliminate substance interference with signal and potentiostat increases the linear detection range of sensor. Non-ideal approach for continuous monitoring because electrochemical sensors require high sampling frequency and destructive analysis, depleting glucose supply. Subject to electrode fouling.	2	Mross <i>et al.</i> [47]
	External	Optical	Mid-infrared (MIR) spectroscopy sensor for glucose concentration analysis	MIR spectroscopy non-invasive sensing is an ideal approach to continuous manufacturing. Further optimization of sensing technique is necessary to introduce system for long-term use.	5	Lüder <i>et al.</i> [48]

Table 2 (Continued)

Analyte	Position	Type	Sensor Name/Description	Potential Use for Cont. Manufacturing	Rank (1–5)	References
	External	Optical	Affinity-viscometry-based micro-sensor probe for continuous monitoring	Probe functions based on noncovalent interactions of glucose with ConA and/or dextran and the resulting change in viscosity of the test solution. It is a nondestructive, invasive measurement techniques. Viscosity of test solution is measured using a microelectromechanical system; however, the probe lacks robustness because the accuracy changes depending on the bioprocess growth medium. Further development of technology is essential for it to be inherently applicable to continuous bioprocessing.	2	Theuer <i>et al.</i> [27]
	External	Optical	MIR spectroscopy and Multivariate Data Analysis (MVDA) predictive modeling	MIR spectroscopy non-invasive sensing is an ideal approach to continuous manufacturing. MVDA improves the robustness of the system. Further optimization of sensing technique is necessary to utilize system for long-term use.	5	Lüder <i>et al.</i> [48]
	External	Optical	MIR sensor using the attenuated total reflection (ATR) technique	Non-invasive MIR spectroscopy sensing is an ideal approach to continuous manufacturing. ATR is a multi-channel midinfrared sensing technique that reduces cost, size and simplifies operability from conventional FT-MIR. Further optimization of sensing technique is necessary to utilize system for long-term use.	5	Schalk <i>et al.</i> [49]
	External	Optical	Microfluidic device with integrated silicon photodiode for chemiluminescence detection of glucose	Chemiluminescence is an alternative to amperometric detection. This approach requires the implementation of fresh enzymatic solution with each use over a course of several days. The device maintenance and destructive analysis of this technique do not warrant it as a useful application for continuous analyte monitoring.	1	Gruber <i>et al.</i> [25], Moon <i>et al.</i> [50]

Table 2 (Continued)

Analyte	Position	Type	Sensor Name/Description	Potential Use for Cont. Manufacturing	Rank (1–5)	References
Glucose, Lactate	Internal	Electrochemical	Active Radio Frequency Identification (RFID) sterile glucose and lactate sensor , capable of up to 14 days of continuous monitoring	Stable method of detection with 0.1 mmol limit of detection. Not useful for PSD culture due to destructive analysis of analyte. Subject to electrode fouling.	3	Kuhner [23]
	External	Electrochemical	Transparent, modular, hybrid microsystem based on glass chip . Combines a cell cultivation chamber and microfluidics with chemo- and biosensors.	Not useful for PSD culture due to destructive analysis of analyte. Subject to electrode fouling. No long-term stability data available in cell culture.	2	Weltin <i>et al.</i> [51]
	External	Electrochemical	Multisensor silicon chip with four individual amperometric sensor elements: two concentric three electrode configurations for a glucose and lactate sensor, interdigitated electrodes for cell density, and a pH sensor	Not useful for PSD culture due to destructive analysis of analyte. Subject to electrode fouling. No long-term stability data available in cell culture.	2	Mross <i>et al.</i> [52]
	External	Optical	Spectrometer used to make Near-infrared (NIR) measurements on medium passing through flow cell-coupled with bioreactor.	Non-destructive, noninvasive and robust sampling technique. The volume of the flow cell is not specified, and the sensor loses accuracy as cell density increases. Flow cell requires additional steps to ensure sampling technique does not contaminate the bioprocess. Optimization of long term parameters for culture monitoring is necessary to guarantee reliable results.	4	Qiu <i>et al.</i> [53]
Glucose, Lactate, Glutamine	External	Electrochemical	A polymer laminate technology-based microsystem comprising of integrated pre-calibrated electrochemical biosensors for monitoring of lactate, glucose, glutamine, and glutamate in a flow-through cell	Direct and simultaneous monitoring technique that has long-term operational stability (minimum 1 week). Simple, robust and small size with wireless data transfer. Flow cell only requires 150 nL volume of culture so it can minimize destructive analysis. Disposable technology designed for use in disposable bag reactors. High selectivity and specificity for analytes.	4	Moser & Jobst [54]
Glucose, Lactic Acid	External	Optical	NIR spectroscopy used to take measurements in a flow cell that is inserted in a closed loop having culture circulated	Noninvasive, rapid, non-destructive technique; High accuracy, precision and good stability (up to 34 hours continuous monitoring in a continuous manufacturing system). Technical aspects (<i>in-situ</i> probe versus external circulation loop used) still need be addressed.	5	Tamburini <i>et al.</i> [55]

Table 2 (Continued)

Analyte	Position	Type	Sensor Name/Description	Potential Use for Cont. Manufacturing	Rank (1–5)	References
Glucose, Glutamate, Glutamine	External	Electrochemical	Silicon-based microfluidic amperometric enzyme biosensor chip coupled to flow injection analysis.	Cost effective, small size analysis tool with specificity and selectivity for analytes using enzymatic based sensors that show 9-day stability and are comparable to commercially available instruments. Although only offline testing performed, flow injection setup can facilitate an automated sampling loop from bioreactor making more ideal for continuous monitoring. Potential for destructive analysis in small volumes due to analyte consumption.	3	Bäcker <i>et al.</i> [24]
Glucose, Glycerol, Ethanol, Acetate, Phosphate, Ammonium	External	Optical	Online sampling and analysis device with an automatic photometric robot. The sampling system includes a ceramic filtration probe, a transport line, a controlled pump and a self-constructed cuvette	Uses a self-constructed flow through cuvette filled with glass beads to take on-line measurements of analyte concentrations. Sampling volumetric flow rate is 1 mL/min in a 1 mL cuvette, which may not be ideal for PSD systems. Calculates concentrations from automatically diluted samples and commercially available photometric assays. Can be used with different types of commercially available assays.	4	Dietzsch <i>et al.</i> [28]
Lactate	External	Electrochemical	Self-powered amperometric lactate biosensor based on lactateoxidase immobilized in dimethylferrocene-modified LPEI.	Tested thoroughly as implantable sensor in biomedical applications; however, extensive optimization required for cell culture application. Although enzymatic sensor has selectivity and sensitivity (up to 10 mM) for lactate and the immobilization technique facilitates a self-powered device, there is potential for destructive analysis in smaller volume culture systems.	1	Hickey <i>et al.</i> [56]
Protein Product	External	n/a	Reversed-Phase High-Performance Liquid Chromatography – separation of molecules based on hydrophobicity	<i>Large scale device with high stability and efficiency. Costly and time consuming. Scale of device is not ideal for small volumes.</i>	2	Carr [57]
Protein Product, Biomass	Mixed	Optical	Dielectric spectroscopy – cell sample acts as a dielectric change in electric current-voltage relation used to extract information from sample	Non-invasive and nondestructive analysis with stability greatly depending on electrode polarization. The detection method requires very sensitive and highly optimized instrumentation. The permittivity has high susceptibility to interference. The limiting factor of this technique is equipment performance.	3	Sven <i>et al.</i> [58], Justice <i>et al.</i> [35], Opel <i>et al.</i> [36]

sensor is not static, allowing it to collect a large dataset from various positions in the culture vessel [13].

An optical pH measurement system that is designed for PSD-scale process development utilizes fluorescent dyes in polymer patches that are illuminated with two colors of LED light, and the ratio of emission wavelength is measured and correlated with pH [14]. Similar technology that takes online samples of cell culture in phenol-red medium has been developed to more precisely quantify the change in characteristic absorption of the medium as its color changes in relation to pH [15].

Dissolved oxygen (DO) sensors

Amperometric sensors have traditionally been used for measuring DO, but miniaturization for PSD-scale development is difficult. The consumption of oxygen during the electrochemical reaction could also potentially impact cell growth [8**].

Takahashi *et al.* developed a system that continuously extracts samples from shaker flask culture to perform online electrochemical DO readings in a closed line separate from the flask without stopping flask shaking [16*]. The electrochemical PATsule sensor described in the pH sensor section can also be used to collect DO measurements [13].

A novel mixed sensor that measures rate of DO diffusion into a closed system connected to an optical sensor sitting outside of the culture vessel was developed by Chatterjee *et al.* Oxygen diffuses through a polymer sampling tube sitting in the culture and is pumped to the sensor, where the initial diffusion rate is measured and linearly correlated with DO concentration [17].

Optical sensor patch technology similar to what was described in the pH sensor section was also developed, where the phase angle between the excitation light and fluorescence emission is measured and correlated with DO concentration [18,19]. The patch can also be placed externally on a gas permeable membrane (where the patch is not contacting the cells) that oxygen in the cell culture liquid can diffuse across for external measurements [20]. These patches were found to be less effective for shaker flask culture with low volume and high shaking frequency compared to a method that involves filling the culture with oxygen-sensitive nanoparticles and using the fiber rod of an optical sensor facing the side of the flask to collect external dissolved oxygen tension (DOT) measurements [21*].

Dissolved carbon dioxide (DCO₂) sensors

The classic method for measuring CO₂ employs the Severinghaus principle, where a potentiometric measurement of the change in pH resulting from CO₂ diffusion into the electrode is made [22]. The circulation direct

sampling and monitoring system described in the DO sensors section can be used with this electrochemical CO₂ sensor [16*].

Effective optical DCO₂ sensors for PSD cell culture scale are largely nonexistent, but the diffusion rate-based method described in the DO sensors section is also capable of measuring DCO₂ in cell culture [17].

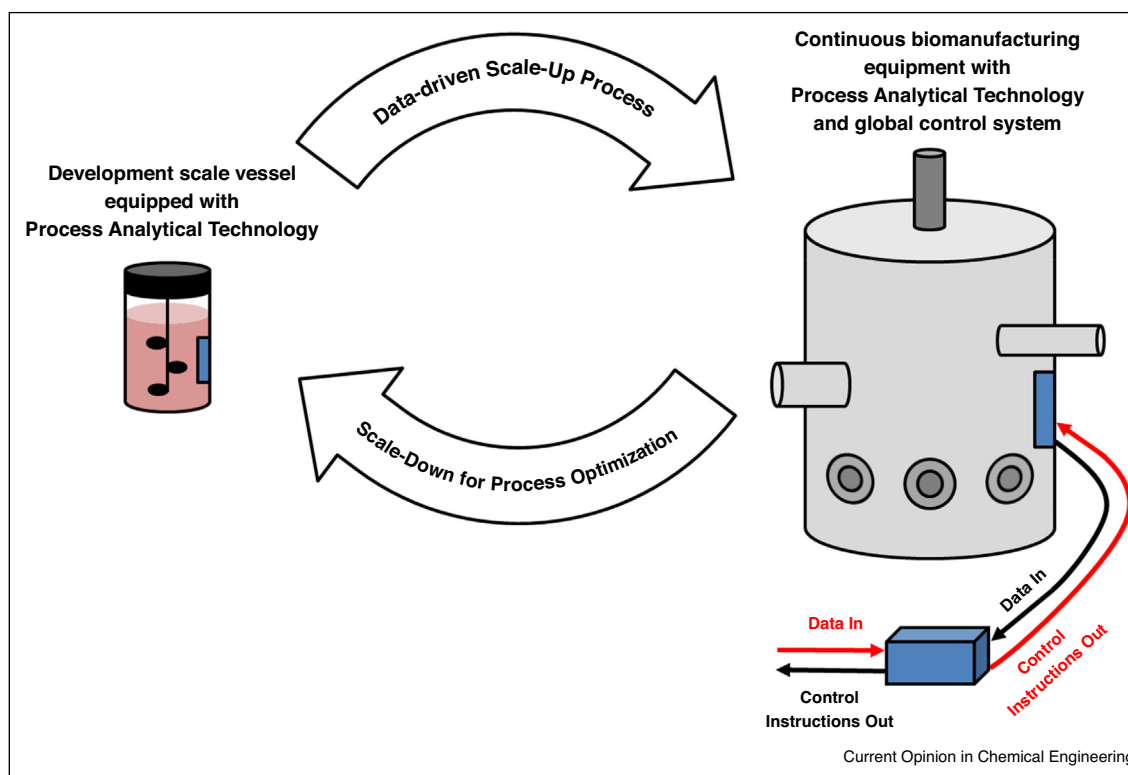
Glucose, glutamine, lactate, ammonia, and protein product sensors

Electrochemical sensors have traditionally been used for these analytes (except for ammonia) and employ one of the following methods of analysis: RFID [23], measurement of enzymatic oxidation products [24], or the use of boronic acids and Concanavalin A or other binding proteins [25]. Current research shows frequent use of enzymatic electrochemical sensors whose highly selective enzymes (i.e. glucose, glutamine, and lactate oxidase) convert the desired analyte into reaction products that generate detectable current. Mross *et al.* developed a device that couples a glucose-oxidase based electrochemical sensor with a CMOS potentiostat and a two-layer membrane containing cellulose acetate and polydimethyl-siloxane (PDMS) [26]. The membrane addition enhances the enzymatic sensor by eliminating interference and reducing glucose flux for detection of a wide range of concentrations.

Optical sensors include different types of spectroscopy, affinity viscometry based on the use of Concanavalin A [27] and a photometric robot for online monitoring [28]. Unlike many traditional optical sensors, spectroscopy methods can detect multiple analytes simultaneously [29]. Tamburini *et al.* applied near-infrared (NIR) spectroscopy, which detects spectra absorption unique to specific analytes, to perform automated, online, external analyte measurements in a continuous bioprocessing setup [30]. The analytes were measured online using a closed-loop pumping system to recirculate medium back and forth from the reactor to the NIR flow cell. Additionally, in-line monitoring using NIR was achieved by immersing a probe, connected to a spectrometer via a fiber, directly into the bioreactor.

Raman spectroscopy operates by detecting inelastic scattering of light that is produced from a laser illumination source, which provides molecular data for the excited molecules in the sample. Using an in-line probe, the first reported real-time measurements of several nutrients and metabolites were measured in CHO cell fermentation of 500 L bioreactors [31]. *In situ* monitoring using a Raman probe in conjunction with chemometric data analysis and partial least squares regression modeling was performed at the 3 L and 15 L scale for detecting multiple analytes [29].

Figure 2



Summary of how process analytical technology in process development will support the implementation of continuous production methods monitored with advanced global control systems in commercial biomanufacturing.

Another spectroscopy method is multi-wavelength fluorescence spectroscopy, which is highly selective and sensitive [32]. A fluorescence-based sensor that was shown to be capable of non-continuous analyte monitoring [32] was later modified into a soft sensor based on an Extended Kalman Filter (EKF) that is capable of data-driven predictions of process data between sample measurements, effectively allowing it to continuously monitor the CHO cell culture process [33]. Soft sensors based on implicit models that assume no variable independence or relationships are useful for bioprocesses because of the system complexity [34*].

Another useful method is dielectric spectroscopy, in which the cell culture acts as a dielectric in an electric field. The change in electric current-voltage relationship is used to extract process information [35]. Dielectric spectroscopy was used with three different data modeling methods to accurately measure viable cell concentration (VCC) and predict data in-between measurements in batch and fed-batch CHO cell culture [36].

Applications in continuous biomanufacturing

For continuous upstream processing of biopharmaceutical products, cells are generally cultured for long periods

of time in perfusion bioreactors that continuously replace culture media in the tank with fresh media and remove used media [37]. This technology continues to be improved through development of more stable cell lines for productive long-term culture and as media and bioreactor conditions are further optimized for higher viable cell density. The first pieces of downstream unit operations for continuous manufacturing have been developed in recent years and need significant evaluation and testing before being implemented at the commercial scale [1]. The sensors discussed in this review will be useful in scaled-down process characterization of these novel unit operations and processes, and some of them could be used in commercial scale production in the future [5].

Although PAT has become much more sophisticated in recent years for real-time data monitoring and is being developed for different scales of process development as described here, most of the current technology is not powerful enough to control a continuous manufacturing process. However, several pieces of technology discussed in this review can provide useful data in continuous manufacturing process development and could someday be implemented in fully integrated commercial-scale production. The best performing sensors that can be used

at different scales were ranked highly in Tables 1 and 2 based on the criteria mentioned previously. Sensors that are only effective at a specific scale and/or have potential issues that need to be addressed before using them in continuous bioprocesses were ranked with a 3 or 4. Soft sensors and other spectroscopy methods combined with predictive data modeling would especially be useful with an integrated control system at the commercial scale because they could predict process changes and modify operational conditions preemptively. The PATsule for dynamic monitoring of pH and DO throughout the cell culture would be able to provide robust data throughout the continuously mixed perfusion bioreactor that could be transmitted to an external control system.

Developing a strong understanding of a new continuous production process at the development scale using the technology described here will be crucial before fully continuous manufacturing facilities can be used for commercial production. Real Time Release Testing (RTRT) that ensures product quality based on real-time process data is also supported by the robust process characterization with these sensors and similar technology used at the commercial scale. Product quality can be further validated through emerging rapid quality control technology, such as a microfluidic device that can quickly detect microbial detection in cell culture in less than 30 min [38].

Conclusions

Despite the issues that will need to be addressed before continuous biomanufacturing equipment and processes can be utilized in commercial-scale production, this production strategy has several major benefits over traditional batch production. However, robust process development is necessary for effective implementation of new and emerging continuous manufacturing technologies. Using the sensors described in this review and new technologies being developed will play an important role in characterizing continuous processes and implementing them in commercial-scale production as shown in Figure 2.

Just as many other manufacturing industries have eventually adopted continuous production at the commercial scale, the biopharmaceutical industry is trending toward using this technology [2]. Biomanufacturing is undoubtedly more regulated than most other types of production, but it is important for this industry to evolve and improve as the demand for biopharmaceutical products increases and the global population grows.

Conflict of interest statement

Nothing declared.

References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- of special interest
 - of outstanding interest
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